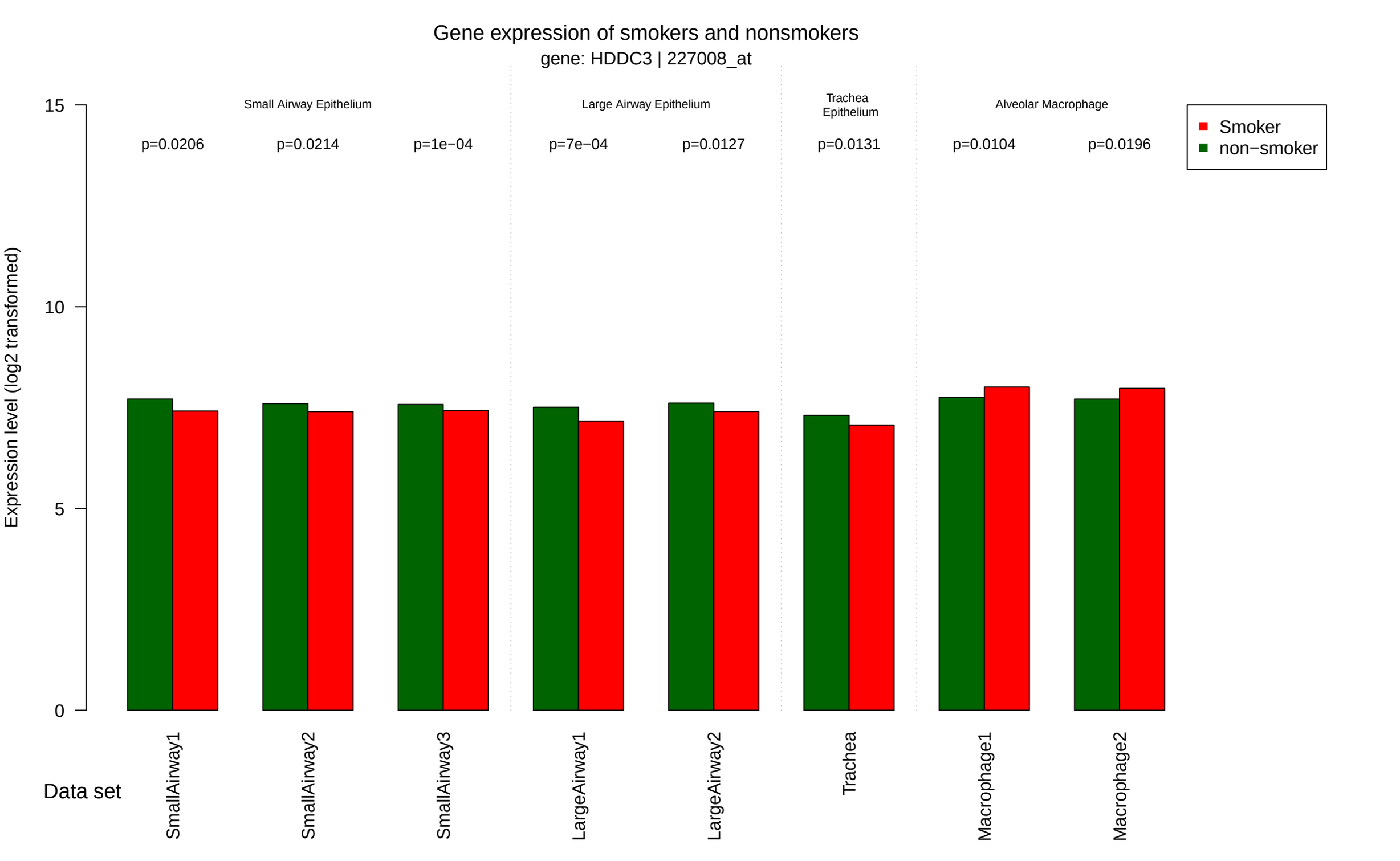
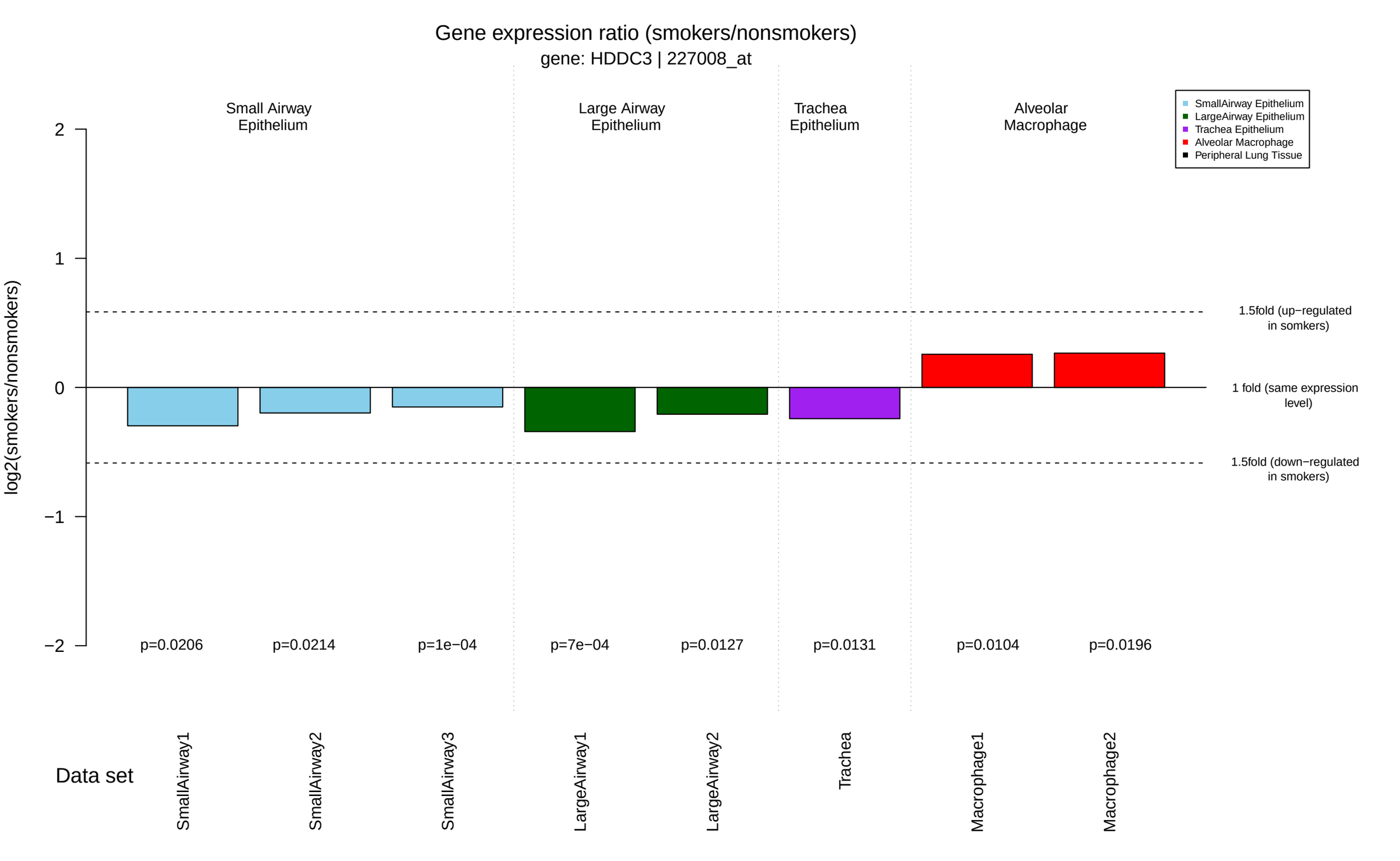
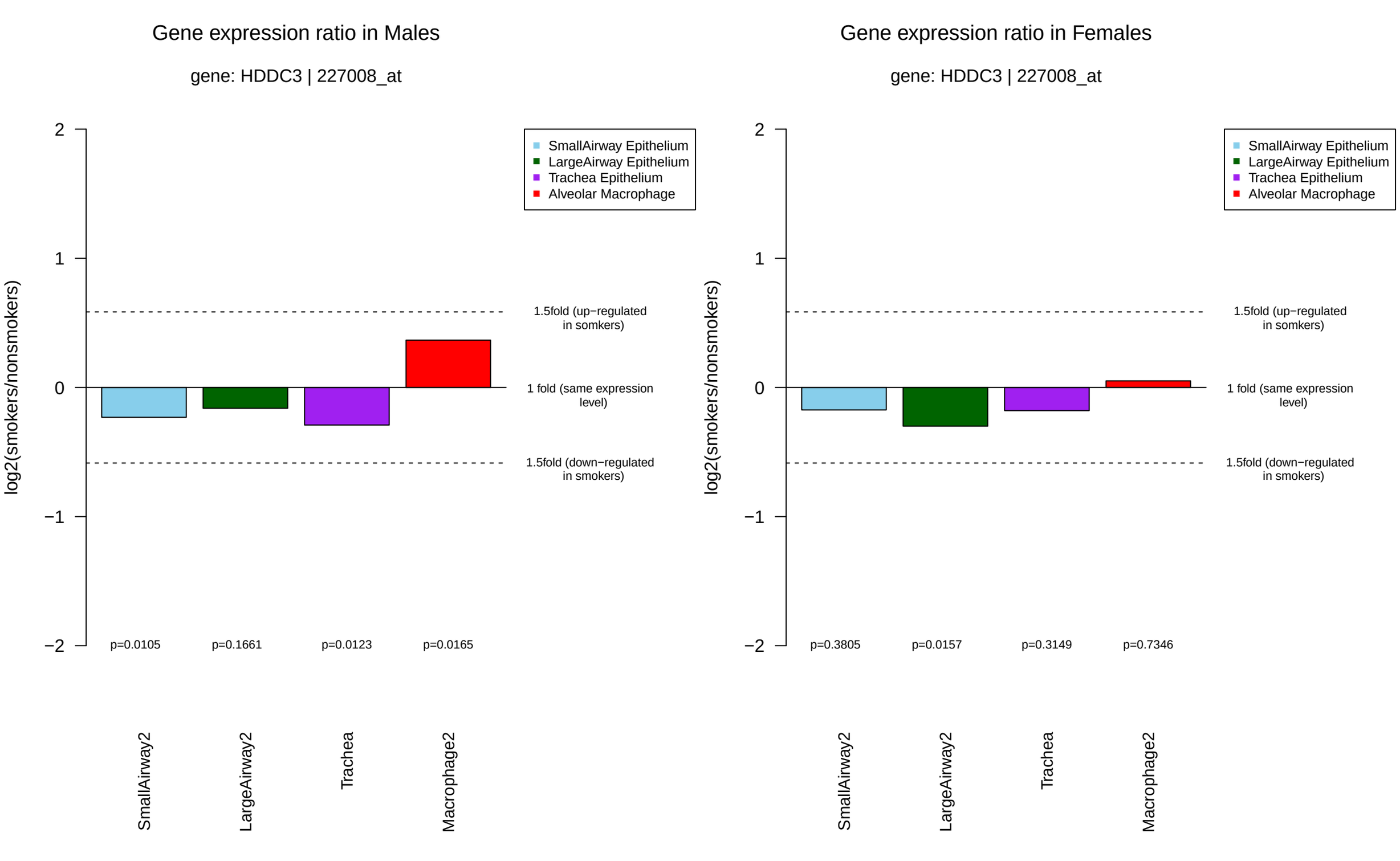
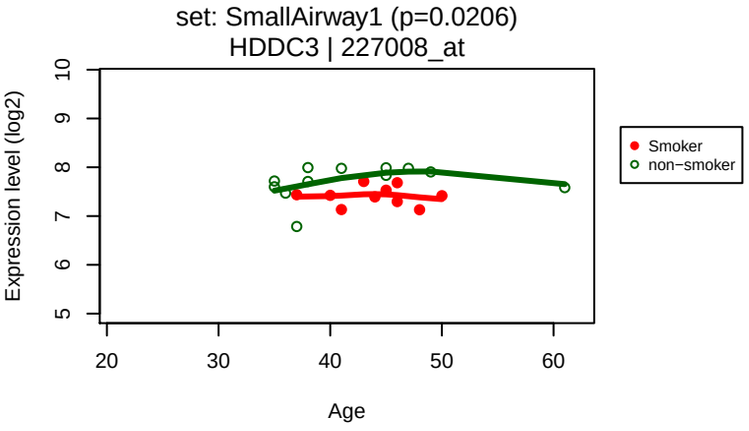
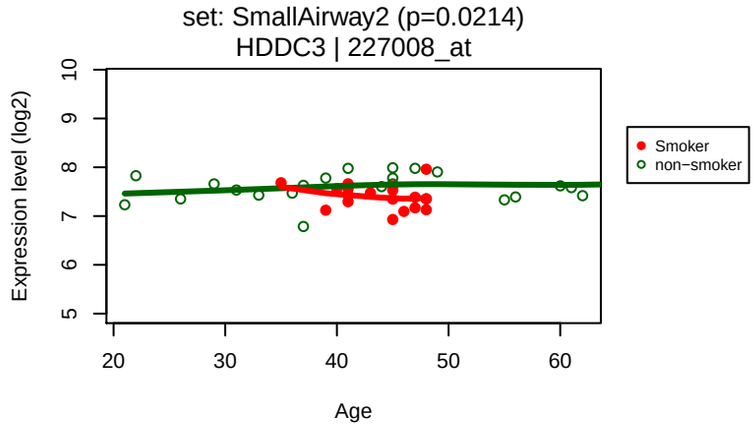
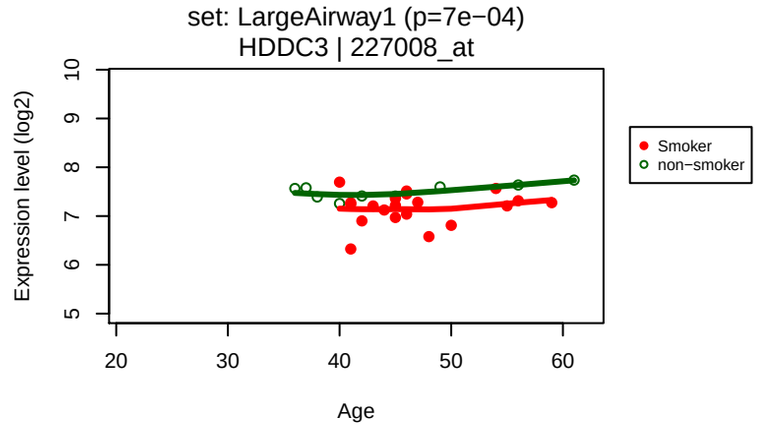

Gene expression of smokers and nonsmokers
gene: HDDC3 | 227008_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0206
p=0.0214
p=1e−04
p=7e−04
p=0.0127
p=0.0131
p=0.0104
p=0.0196
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

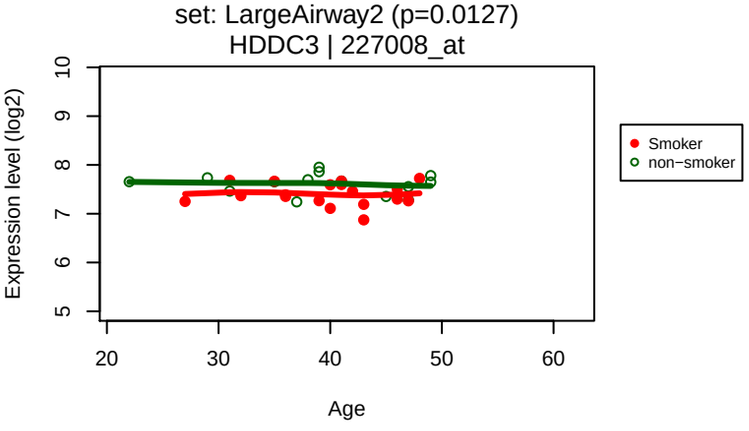

Gene expression ratio (smokers/nonsmokers)
gene: HDDC3 | 227008_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0206
p=0.0214
p=1e−04
p=7e−04
p=0.0127
p=0.0131
p=0.0104
p=0.0196
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

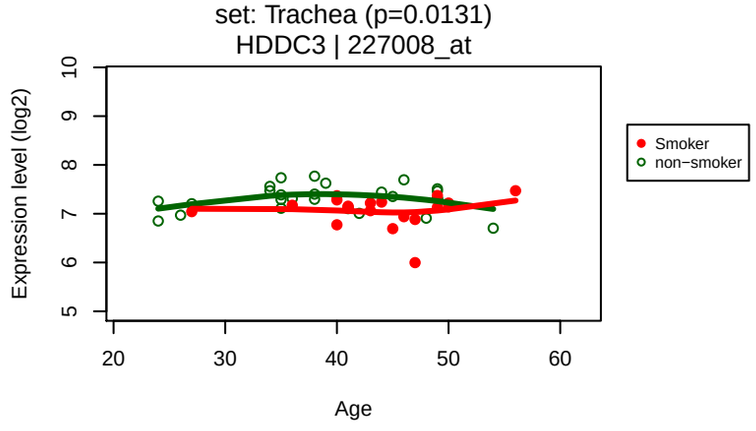

Gene expression ratio in Males
gene: HDDC3 | 227008_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0105
p=0.1661
p=0.0123
p=0.0165
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: HDDC3 | 227008_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3805
p=0.0157
p=0.3149
p=0.7346
SmallAirway2
LargeAirway2
Trachea
Macrophage2

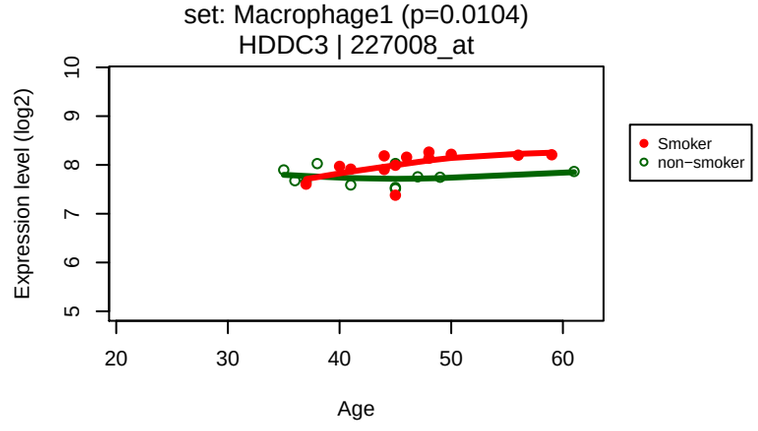

set: SmallAirway1 (p=0.0206)
HDDC3 | 227008_at
Expression level (log2)
Age
Smoker
non−smoker

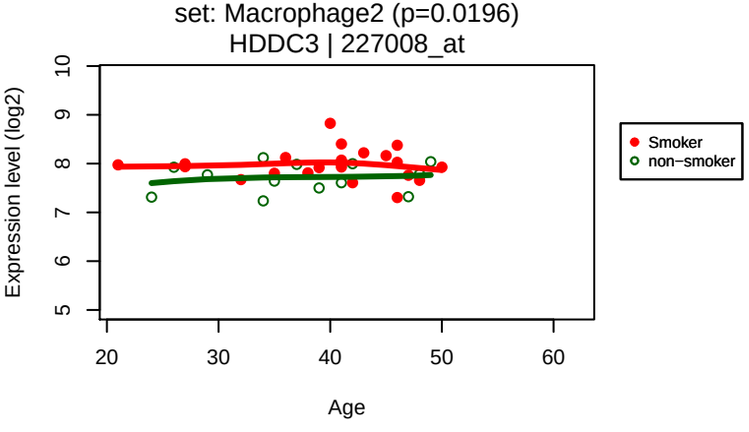

set: SmallAirway2 (p=0.0214)
HDDC3 | 227008_at
Expression level (log2)
Age
Smoker
non−smoker

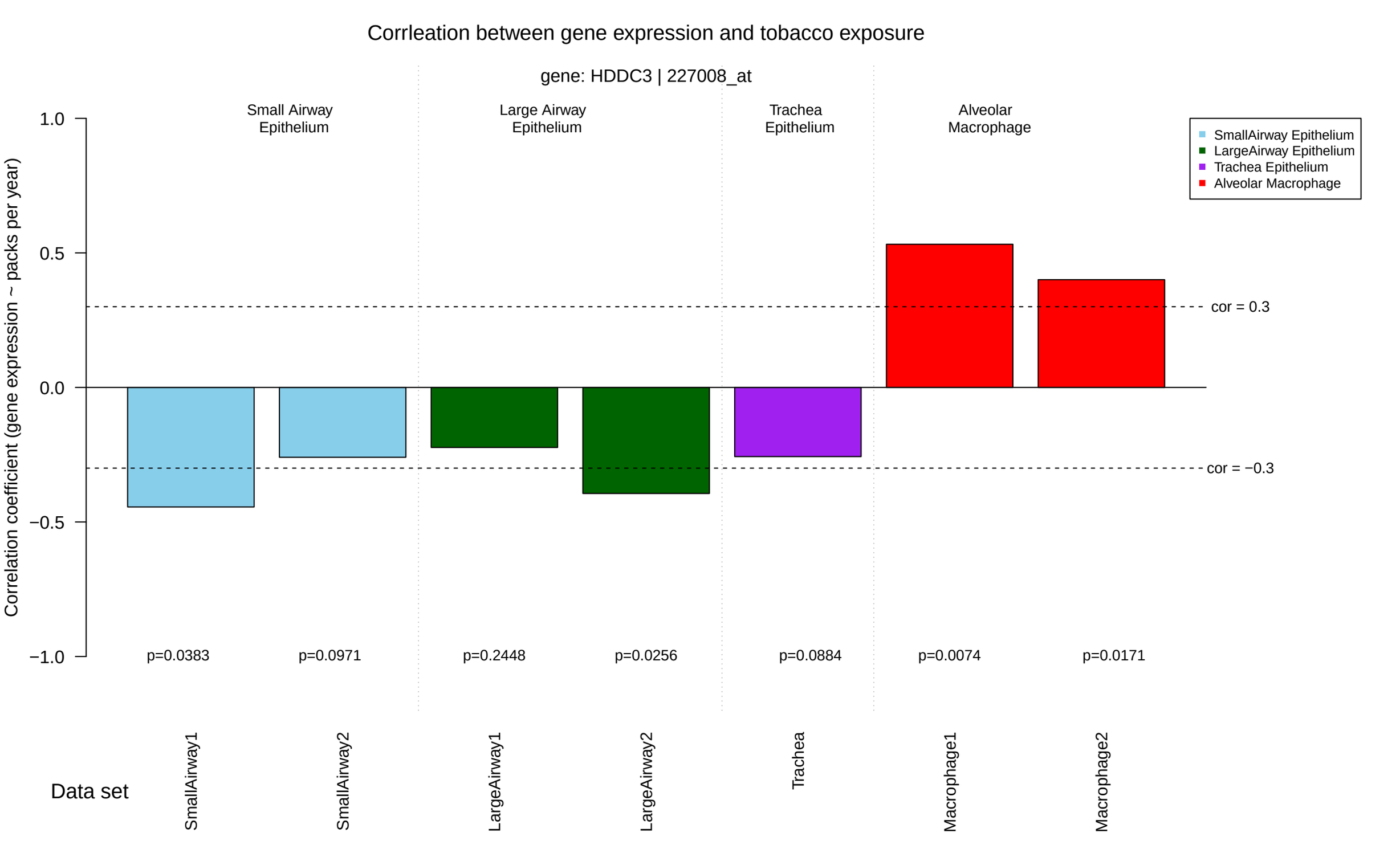

set: LargeAirway1 (p=7e−04)
HDDC3 | 227008_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0127)
HDDC3 | 227008_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0131)
HDDC3 | 227008_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0104)
HDDC3 | 227008_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0196)
HDDC3 | 227008_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: HDDC3 | 227008_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0383
p=0.0971
p=0.2448
p=0.0256
p=0.0884
p=0.0074
p=0.0171
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2